Self-Powered Omnidirectional Pressure Sensing Smart Ball for Hand Rehabilitation Monitoring

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관절 손 재활 모니터링을 위한 자가 구동 전방위 압력 감지 스마트 볼

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Abstract

In this work, we present PEDOT:PSS-doped bifunctional organohydrogel (PPBOH) and silicone-based self-powered pressure-sensing stress ball (SPSB) designed for comprehensive, direction-independent monitoring of complex hand exercises. Incorporating large semiconducting polymer PEDOT:PSS into organohydrogel significantly enhanced its tribopositivity, electrical conductivity, and mechanical robustness while maintaining compressibility, resulting in two-fold increase power density (1.7 W/m^2). Leveraging intrinsic shape adaptability of PPBOH and facile molding approach, conical arrays were molded on PPBOH sphere, encapsulated in electronegative silicone shell, and shielded with outer PPBOH layer. The resulting SPSB exhibits superior omnidirectional sensitivity (0.453 V/N, 2.8-40N; 0.069 V/N, 40-90N). The SPSB was successfully employed for real-time monitoring of comprehensive hand-rehabilitation exercises for multiple users. Collectively, a portable, omnidirectional pressure-sensing ball has been developed, marking substantial advance toward continuous, home-based neuromuscular rehabilitation monitoring.

Keywords: ***Triboelectric nanogenerator; Self-powered pressure sensor; PEDOT:PSS doped organohydrogel; omnidirectional pressure-sensing ball.***

1. Introduction

Hand rehabilitation is a critical component of post-injury and post-surgical management, aimed at restoring hand strength, range of motion, dexterity, and sensory function through evidence-based therapeutic interventions. Effective rehabilitation facilitates functional recovery, prevents secondary complications, and improves overall functional outcomes and quality of life. Several robotic-assisted rehabilitation has emerged as a promising supplement to conventional therapy by enabling intensive, repeatable, and quantifiable training with real-time feedback. However, most existing systems are bulky, costly, and unsuitable for fine, task-specific hand movements. Similarly, clinical tools such as the Jamar hydraulic dynamometer are restricted to laboratory use, lack sensitivity to subtle force variations, and exhibit operator-dependent variability. Recent advances in wearable sensing technologies including capacitive, piezoresistive, piezoelectric, and triboelectric sensors have enabled portable hand function monitoring [1]. Nevertheless, many reported designs are limited by unidirectional sensing, rigid form factors, or external power requirements. Triboelectric nanogenerator (TENG)-based sensors are particularly attractive due to their self-powered operation, flexibility, and biocompatibility. Yet, conventional TENG materials often lack stretchability and 3D adaptability, while gel-based TENGs suffer from reduced output due to adhesive surfaces [2]. Therefore, there is a strong need to develop multifunctional hydrogels that offer minimal adhesive properties as a triboelectric layer with combined aforementioned mechanical and electrical features.

Herein, an omnidirectional self-powered pressure sensing smart ball (SPSB) is developed for hand rehabilitation monitoring using a biocompatible PEDOT:PSS-doped gelatin organohydrogel (PPBOH) and silicone elastomer. The incorporation of PEDOT:PSS enhances the tribopositivity and conductivity while maintaining mechanical compliance. The SPSB consists of a tribopositive PPBOH spherical core with a micro-conical array encapsulated in tribonegative silicone rubber and an outer PPBOH shielding layer to minimize skin-induced triboelectric interference. Contact-separation between the PPBOH microcones and silicone layer during squeezing and releasing generates voltage proportional to applied force from all directions. The optimized structure exhibits stable, reliable, and omnidirectional pressure sensing, demonstrating strong potential making it suitable for hand rehabilitation monitoring.

2. Result and Discussion

The schematic illustration of the omnidirectional self-power pressure sensing smart ball for hand rehabilitation monitoring is shown in **Figure 1a**. The schematic of the working mechanism of the self-powered strain sensor is shown in **Figure 1b**. The omnidirectional cyclic pressing and releasing operations result in the triboelectric voltage based on the contact electrification and electrostatic induction effects induced at the PPBOH cone and negative silicone layer interface, driving the electron flow for triboelectric signal generation. **Figure 2a** shows the structure of a flexible PPBOH/silicone TENG and its electrical performance,

highlighting the influence of PEDOT:PSS concentration on the output voltage and current (**Figure 2b, 2c**), as well as the load-dependent power behavior, achieving a maximum power of 0.681 mW, and a 1.77-fold improvement in peak and average power density after the addition of 1 wt.% PEDOT:PSS. Utilizing the exceptional electrical and mechanical properties of PPBOH, SPSB was developed. The fabrication sequence is shown in **Figure 3a.** The force response of ball was studied over a wide range of force as shown in **Figure 3b,** while force response along all directions demonstrated similar sensitivity (**Figure 3c**), illustrating omnidirectional behavior of the SPSB. Finally, the SPSB was successfully employed for real-time monitoring of comprehensive hand-rehabilitation exercises across multiple users as demonstrated in **Figure 4a-e**.

3. Conclusion

In summary, a portable self-powered smart ball was developed using a PEDOT:PSS-doped gelatin organohydrogel and silicone elastomer. Doping with PEDOT:PSS significantly improved the triboelectric performance, conductivity, and mechanical strength, achieving a open-circuit voltage of 621 V, short-circuit current of 20 μA and a peak power density of 1.703 W/m² nearly twice that of pristine gelatin hydrogel. The smart ball, fabricated through simple 3D molding and silicone encapsulation, exhibits high sensitivity to omnidirectional forces with stable and repeatable outputs. It was successfully used to monitor hand rehabilitation exercises, demonstrating strong potential for portable digital health monitoring applications.

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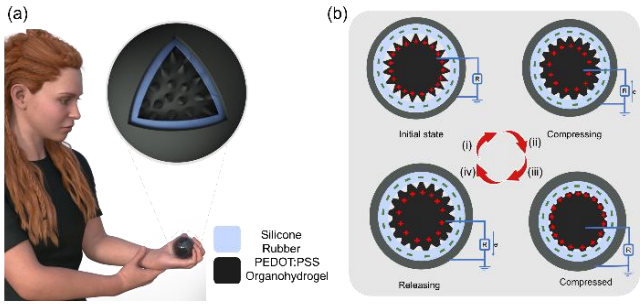

Fig. 1. (a) Schematic illustration of omnidirectional pressure sensor for hand rehabilitation monitoring. (b) Working mechanism of the sensor.

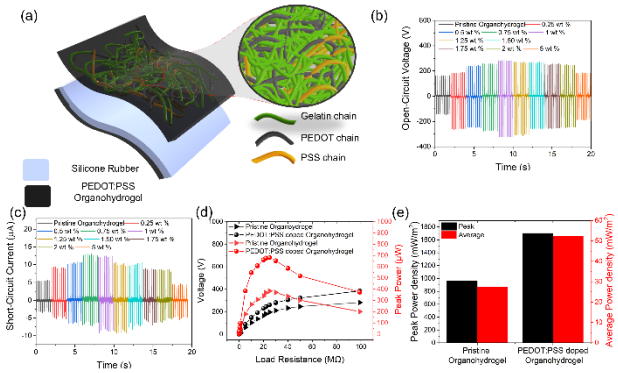

Fig. 2. (a) Schematic diagram of flexible PPBOH and silicone-based triboelectric nanogenerator (b) Open-circuit voltage and short-circuit current with different concentration of PEDOT:PSS in PPBOH. (d) Load voltage and peak power curve of TENG at varying load conditions. (d) Comparison of peak and average power density before and after introduction of 1 wt.% PEDOT: PSS.

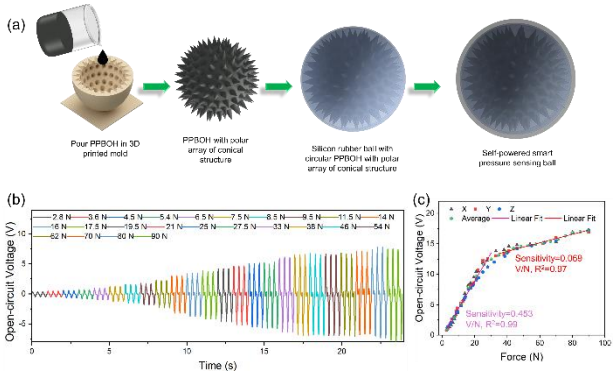

Fig. 3. (a) Fabrication sequence of SPSB. (b) Open-circuit voltage response of SPSS with respect to applied force. (c) Linear response of SPSS along all three primary directions.

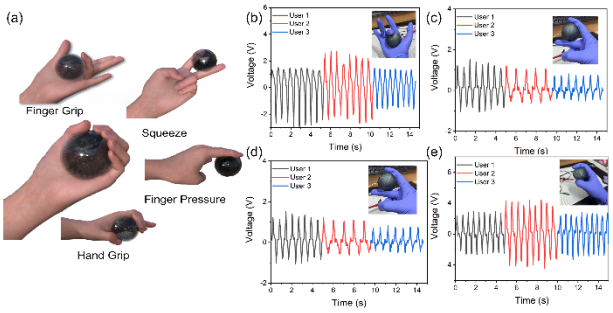

Fig. 4. (a) Schematic illustrations of different hand rehabilitation exercises. Voltage waveform of 3 users while performing (b) finger grip,(c) squeeze, (d) finger pressure, and (e) hand grip.

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